What is turbulence?

Key features, examples, and why it matters

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IBFE Tutorials

A working viewpoint

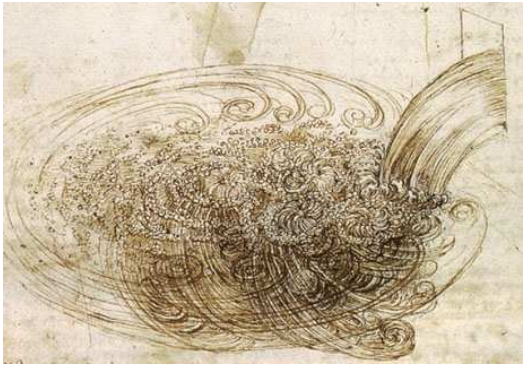
Turbulence is recognized through a collection of dynamical signatures, not one universal checklist.

Adapted from a lecture by Steven L. Brunton.

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

Why turbulence fascinates us

- Turbulence shows up in **art** and **language**: roiling, churning, turbulent.
- We have strong intuition for "turbulent" vs "not," but a single universal definition is hard.
- So: we focus on **key features** and **canonical examples**.



Leonardo da Vinci, *Studies of water*, c. 1510–1512 (Royal Collection, public domain).

Turbulence in art: Van Gogh



Vincent van Gogh, *The Starry Night* (1889), MoMA (public domain).

A practical "fingerprint" of turbulence

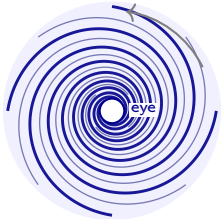
A useful operational fingerprint is that turbulent flows are typically:

1. **Three-dimensional** (3D vortical motion),
2. **Multi-scale** in space and time (big eddies $\rightarrow$ small eddies),
3. **Unsteady** (time-evolving),
4. **Mixing** and strongly transporting scalar/particles.

Mental experiment: add dye upstream $\Rightarrow$ rapid stretching and folding $\Rightarrow$ enhanced mixing.
This is a diagnostic, not a formal definition.

A note on 2D vs 3D turbulence

- In 3D turbulence, kinetic energy is typically transferred toward smaller scales and ultimately dissipated by viscosity.
- Idealized 2D turbulence has two inviscid quadratic invariants: energy tends upscale, while enstrophy tends downscale.
- Rotating or stratified geophysical flows can be **quasi-2D**; finite thickness, friction, and wave dynamics modify the ideal picture.

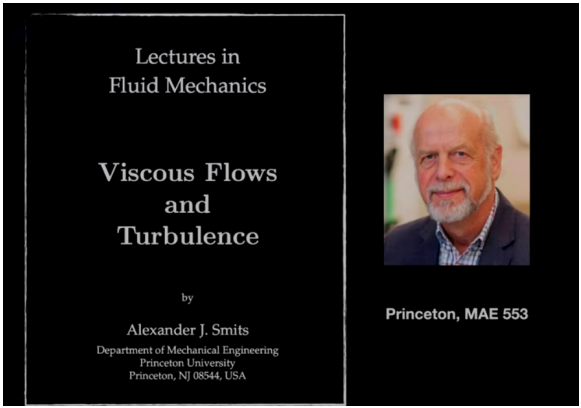


cyclone: $L \sim 500$ km

Schematic (instructor's course notes).

A recommended reference (from the lecture)

- Lex Smits (Princeton): **MAE 553 Viscous Flows and Turbulence** notes.
- Use as a companion source for canonical flows and experimental intuition.



Title page of A. J. (Lex) Smits, *MAE 553: Viscous Flows and Turbulence* lecture notes (Princeton University).

Example: a turbulent boundary layer (simulation)

- High-fidelity simulation of a developing boundary layer (think: flow over a wing/car).
- Observe: **3D**, **unsteady**, **multi-scale**, **mixing**.

3D Simulation (YouTube)

Hallmark: multi-scale structure

- "Zoom in anywhere" $\Rightarrow$ you see more structure.
- Multi-scale content is a quick visual diagnostic of turbulence.

Fun aside (from lecture): blockbuster explosions look "high budget" when they show many scales.



Still from *Oppenheimer* (dir. C. Nolan, Universal Pictures, 2023); shown for educational commentary, rights reserved by the studio.

Turbulence at planetary scales

- Ocean currents and atmospheric patterns are highly turbulent.
- Turbulent transport $\Rightarrow$ global mixing of heat, salt, nutrients, pollutants.



Photo: Kent Smith, Gulf of Alaska (2007), glacial meltwater meeting ocean water.

Rayleigh–Bénard convection and chaos

- Heated from below, cooled from above $\Rightarrow$ convective instability.
- Unsteady + multi-scale + mixing (often used to motivate chaotic dynamics).
- Historical link: Lorenz (1963) studied simplified models of convective rolls.

3D Simulation (YouTube)

Mathematical entry point: incompressible Navier–Stokes

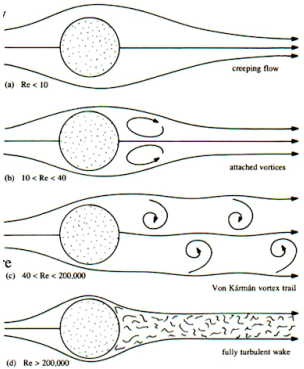
$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}} \Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

$$\mathbf{u}(\mathbf{x}, t) = \begin{bmatrix} u(\mathbf{x}, t) \\ v(\mathbf{x}, t) \\ w(\mathbf{x}, t) \end{bmatrix}$$

- $\boldsymbol{u}(\boldsymbol{x}, t) \in \mathbb{R}^3$ is a 3-component velocity field.
- Rich behavior emerges as parameters/forcing/geometry change.

Reynolds number: a control knob for complexity

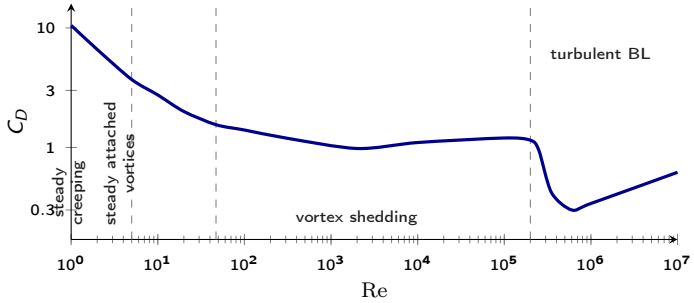
- For chosen characteristic scales U and L, the **Reynolds number** compares inertial and viscous effects:
$$\mathrm{Re} \sim \frac{UL}{\nu}.$$
- It is a **control parameter**, but geometry, forcing, roughness, and disturbances also determine the observed state.
- In high-Re, 3D, locally isotropic turbulence, Kolmogorov scaling estimates $L/\eta \sim \mathrm{Re}_L^{3/4}$; this is not a universal one-to-one law.



Flow regimes past a cylinder; after S. Vogel, *Life in Moving Fluids*, 2nd ed. (Princeton Univ. Press, 1994) (figure number to confirm).

Drag depends strongly on Reynolds number

- As flow regime changes with Re, the drag coefficient can change dramatically.
- Implication: "just changing speed" can change force laws in big ways.



Drag coefficient of a circular cylinder versus Reynolds number with flow regimes. Schematic (instructor's course notes).

Aside: Galileo's ball drop through multiple regimes

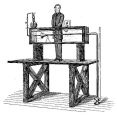
- Falling ball accelerates $\Rightarrow$ effective Re increases during the fall.
- Drag force can be highly unsteady and regime-dependent.
- Heavy objects: turbulent drag may be negligible relative to weight; light objects: not.
- The gravitational field g may be nearly constant, while the ball's acceleration need not remain g once drag is appreciable.



Portrait of Galileo Galilei (public domain); the legendary experiment at Pisa.

Why "Reynolds"? Pipe-flow experiments (1883)

- Osborne Reynolds observed changes in pipe flow as speed and disturbance conditions varied:
  - laminar (layered, steady),
  - transitional,
  - turbulent (roiling, disordered).
- The laminar solution exists at every Re; pipe transition is subcritical and depends on finite-amplitude disturbances.
- Pipe flow remains a core engineering problem: pumping water, sewage, and process fluids efficiently.
- Textbooks quote $\mathrm{Re}_D \approx 2300$ for the onset of transition in pipes; Reynolds (1883) found transition near $\mathrm{Re} \approx 13{,}000$ with a disturbed inlet and could keep flow laminar to far higher Re with care. Deck 5-4 discusses the modern value $\mathrm{Re}_D \approx 2040$.



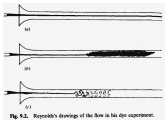


Fig. 9.2. Reynolds's drawings of the flow in his dye experiment.

Apparatus and dye-filament plate from O. Reynolds, *Phil. Trans. R. Soc. Lond.* **174** (1883), 935–982 (public domain).

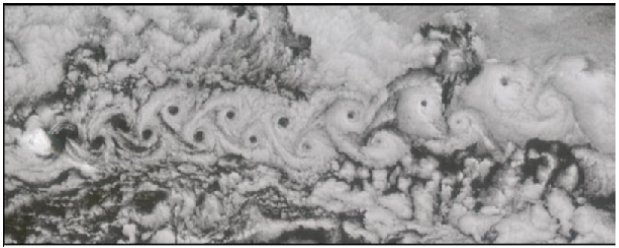
Transition to turbulence

- A classical experiment is to inject dye into a tube to visualize the flow.
- As Re and/or disturbance amplitude increase, the dye filament may become unsteady and then turbulent.
- A single critical value is not universal: inlet noise, roughness, and protocol matter.

Pipe flow transition to turbulence (YouTube)

Turbulence is not "random": coherent structures

- Turbulence has stochastic aspects, but it is not structureless noise.
- **Coherent structures** (organized patterns) can persist within complex flows.
- Similar large-scale patterns can appear across wildly different Reynolds numbers.

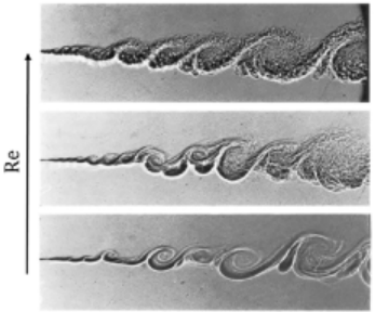




M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982).

Mixing layers: laminar vs turbulent share patterns

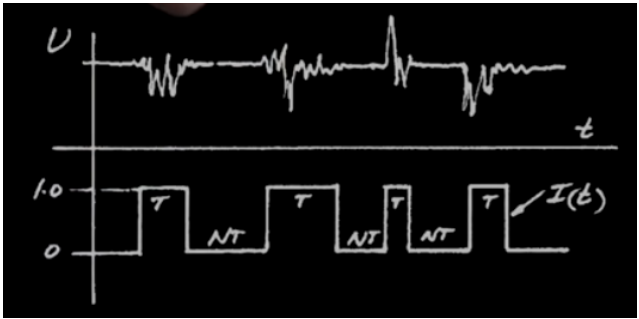
- Classic mixing-layer images show:
  - Bottom: low-Re (laminar),
  - Middle: intermediate-Re (transitional), and
  - Top: high-Re (turbulent).
- All cases share recognizable, coherent structures.
- Turbulence adds scales and distortion, but does not erase organization.



Shadowgraphs of a plane mixing layer at increasing Reynolds number; G. L. Brown & A. Roshko, *J. Fluid Mech.* **64**, 775–816 (1974); reproduced in Van Dyke, *An Album of Fluid Motion.*

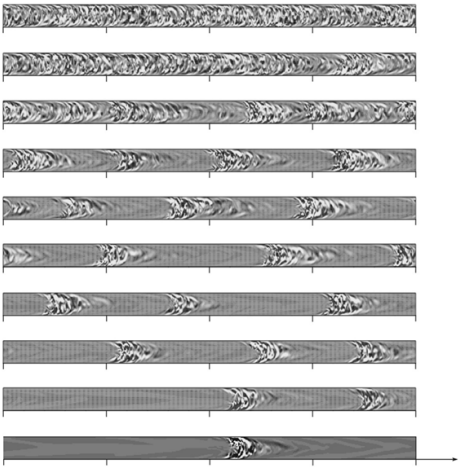
Another hallmark: intermittency

- Turbulence can be **patchy** in space/time:
  - turbulent region $\rightarrow$ calm region $\rightarrow$ turbulent again.
- Picture: smoke plume, clouds in the boundary layer, bursts of turbulence.
- Useful concept: an "intermittency function" that switches on/off turbulence.



Source: Smith and Smits, 1988.

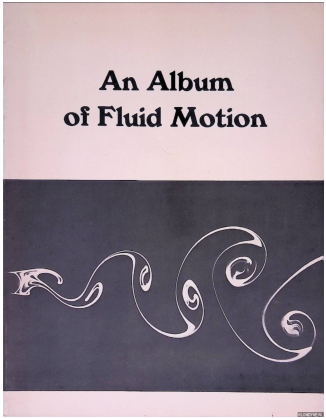
Intermittency in pipe flow



Source: Nikitin, N. & Pimanov, V.. (2015). Numerical study of localized turbulent structures in a pipe. Fluid Dynamics. 50. 655-664.

How we handle complexity: canonical flows

- Canonical flows: isotropic turbulence, wall-bounded flows, boundary layers, mixing layers, pipe flow, channel flow, cavity flow, wakes, jets.
- Strategy: distill turbulence into idealized building blocks.
- Real systems combine many of these (boundary layer + wake + jet + mixing layer).

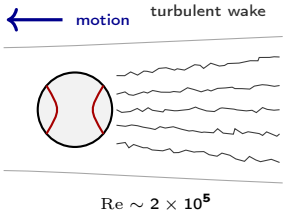


M. Van Dyke, *An Album of Fluid Motion* (Parabolic Press, 1982), cover.

Examples of canonical flows and why they matter

Wakes

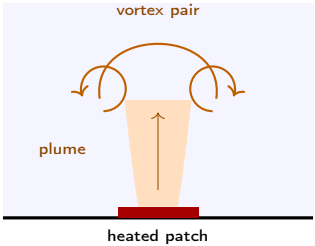
- drag and unsteady forces
- vehicles, trains, projectiles, fish, birds



Schematic (instructor's course notes).

Mixing layers

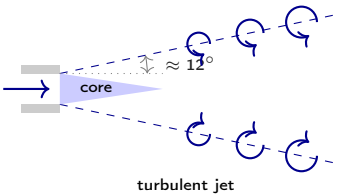
- shear-driven mixing
- ocean waves, atmospheric inversion layers



Schematic (instructor's course notes).

Jets

- thrust and power
- noise and quality-of-life impacts



Schematic (instructor's course notes).

End goal: predict, optimize, and control

- Engineering and scientific aim: develop models that predict how outcomes change with design knobs.
- Examples: increase lift, decrease drag, enhance mixing, reduce noise.
- Control: actively manipulate flow for better performance.



Source: Ball, P. Wing and fin motions share universal principles. Nature (2014).

Preview: Reynolds decomposition

Recall the Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \frac{1}{\mathrm{Re}} \Delta \boldsymbol{u},$$
$$\nabla \cdot \boldsymbol{u} = 0.$$

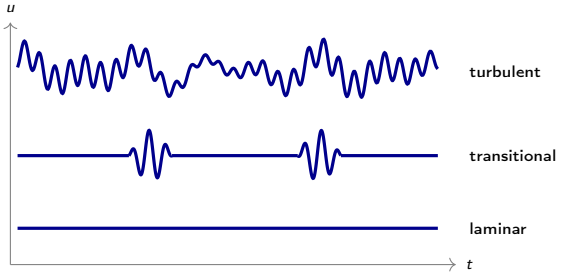
Choose an averaging operator and write the Reynolds decomposition:

$$\boldsymbol{u}(\boldsymbol{x}, t) = \overline{\boldsymbol{u}}(\boldsymbol{x}, t) + \boldsymbol{u}'(\boldsymbol{x}, t), \qquad \overline{\boldsymbol{u}'} = \boldsymbol{0}.$$

- The mean may be an ensemble, spatial, phase, or time average.
- For statistically stationary, ergodic flow, a sufficiently long time average can represent the ensemble mean.
- Fluctuations $\boldsymbol{u}'$: turbulent deviations around the mean.

Why averaging became natural historically

- Before modern field measurements (e.g., PIV/laser sheets), data were often pointwise.
- Hot-wire anemometry: long time series at one point $\rightarrow$ build statistics.
- Averaging is both practical (data) and useful (reduced modeling complexity).



Sketch of laminar, transitional, and turbulent velocity signals recorded at a point (e.g. by a hot-wire anemometer). Schematic (instructor's course notes).

Takeaways

- Turbulence is hard to define, but easy to recognize by its **fingerprint**:

$$\text{3D} \;+\; \text{multi-scale} \;+\; \text{unsteady} \;+\; \text{mixing}.$$

- The Navier–Stokes equations remain the anchor; Re is an important control parameter, not a complete state label.
- Coherent structures and intermittency mean turbulence is **not** pure randomness.
- Canonical flows are how we build understanding and models.
- What's next: Reynolds averaging/closure.

Selected references

- O. Reynolds, "An experimental investigation of the circumstances which determine whether the motion of water shall be direct or sinuous," *Phil. Trans. R. Soc. Lond.* **174** (1883), 935–982. doi:10.1098/rstl.1883.0029.
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